DL Graded Quiz M7

Question 1

Determine the **correct** statement about the Walk Forward method.

- It is similar to an ensemble method where we combine the predictions from models with different configurations.
- It yields a prediction for all the observations in the time series except for those in the initial training sample.
- It is similar to an ensemble method because we generate many predictions for the observations in the test set.
- (◉) It is immune to the leakage problem.

Question 2

Consider a regression problem with a single output to predict. In a mini-batch with two instances, we obtain the following:

y_true= tf.constant([0.5, 1.2, 0.2, 1.2])

y_pred = tf.constant([0.65, 0.43, 0.05, 2.1])

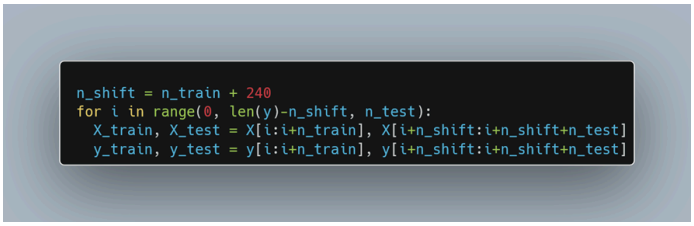
What is the Mean Squared Logarithmic Error loss (average loss over the mini-batch)?

- 0.082
- -0.051
- 0.065
- (◉) 0.183

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Question 3

Consider the following modification to the Walk Forward loop implemented in Lesson 2.



```
n_shift = n_train + 240
for i in range(0, len(y)-n_shift, n_test):
  X_train, X_test = X[i:i+n_train], X[i+n_shift:i+n_shift+n_test]
  y_train, y_test = y[i:i+n_train], y[i+n_shift:i+n_shift+n_test]
```

Determine the right answer.

- It will reduce the likelihood of backtest overfitting by excluding 240 observations where the training sample and the test sample overlap.
- It will reduce the likelihood of backtest overfitting by increasing the size of the training tests by 240 observations.
- It will reduce the likelihood of backtest overfitting by excluding 240 observations from the training set that are contiguous to the test set.
- (●) It will increase the likelihood of backtest overfitting by increasing the size of the training tests by 240 observations.

Question 4

In the Combinatorial Purged Cross-Validation, k refers to which of the following?

- The number of backtest paths that we can generate
- The number of observations that compose the time series
- The number of groups into which we split the time series
- (●) The number of groups that we include in each test sample

Question 5

When should a non-anchored Walk Forward backtest be preferred over an anchored method?

- When the relationship between features and labels seldom changes over time
- When the relationship between features and labels changes a lot over time
- When we want to increase the number of observations in our backtests
- (●) When we want to reduce the number of observations in our backtests

Question 6

Consider a regression problem with a single output to predict. We want to build a custom loss function that computes the Huber loss while training the model using "fit()" in TensorFlow-Keras. We define the following function:

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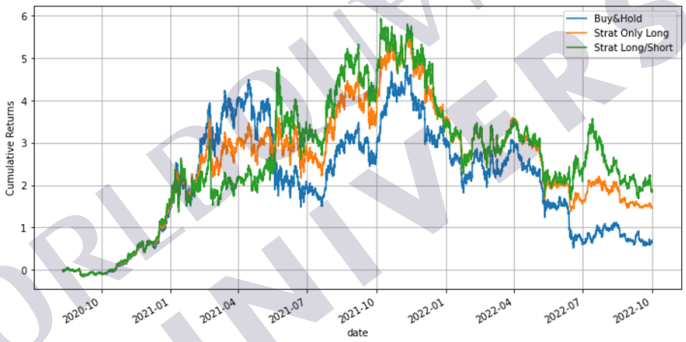
```
def custom_huber(delta):
  def custom_loss(y_true, y_pred):
      error = tf.math.abs(y_true - y_pred)
      ### EMPTY LINE ###
      return loss
  return custom_loss
```

Determine the line of code that should appear on ### EMPTY LINE ### to compute the Huber loss.

- loss = 0.5 * tf.where(delta >= error, tf.square(error), 2*delta*error-delta**2)
- loss = 0.5 * tf.where(delta > error, tf.square(error), 2 * delta + delta * (error - delta))
- loss = 0.5 * tf.where(delta < error, tf.square(error), delta ** 2)
- loss = 0.5 * tf.where(delta < error, tf.square(error), delta ** 2 + delta * (error - delta))

Question 7

A figure of the following type arises from the Walk Forward loop implemented in lesson 2.



Consider the orange line, which plots the cumulative returns from a backtest that takes a long position on Bitcoin when the model predicts a positive return. Choose the right answer.

- The position taken in the strategy arises from the predictions of a single model that uses the preceding 5,000 training observations for each date.
- The position taken in the strategy arises from the prediction of a single model.
- The position taken in the strategy arises from averaging the predictions of the 19 trained models for each test date.
- The position taken in the strategy arises from the predictions of a model that uses 1,000 training observations that are randomly sampled.

Question 8

Consider a time series of 9,000 observations of returns of a security over which we want to perform backtesting of a trading strategy. We use the Combinatorial Purged Cross-Validation method, where we split the data into 7 groups, and each test sample includes 2 groups. How many total backtest paths can we generate?

- 5

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- 4
- 7
- 6

Question 9

Determine which of the following ensemble methods involve training the models of the ensembles on **different** subsets of data.

- Bagging
- Boosting
- Model-Averaging
- Stacking

Question 10

Consider a regression problem with a single output to predict. We want to build a custom loss function that computes the Mean Squared Logarithmic Error while training the model using "fit()" in TensorFlow-Keras. The output layer generates values strictly greater than -1. We define the following function:

```
def custom_loss(y_true, y_pred):
    #### EMPTY LINE ####
    return loss
```

Determine the line of code that should appear on ### EMPTY LINE ### to compute the Mean Squared Logarithmic Error.

- loss = tf.square(tf.math.log(y_true + 1.) - tf.math.log(y_pred + 1.))
- loss = tf.square(tf.math.log(y_pred) - tf.math.log(y_true))
- loss = tf.math.reduce_sum(tf.square(tf.math.log(y_true + 1.) - tf.math.log(y_pred + 1.)))
- loss = tf.math.reduce_mean(tf.square(tf.math.log(y_true + 1.) - tf.math.log(y_pred + 1.)))

Question 11

Determine the **correct** statement about the Walk Forward method implemented in lesson 2.

- It tunes the hyperparameters by evaluating the model performance across test sets.
- It tunes the hyperparameters in each training set using the model performance in the previous training set.
- It trains models in each training set using the trained weights in the previous training set as initial condition.
- It trains models in each training set using random initial weights.

Question 12

Consider a backtest procedure where we drop 10% of the initial observations in the training sets that trail the test sets. This procedure is labeled as which of the following?

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- K-fold Cross-Validation
- Walk Forward
- Purging
- Embargoing

Question 13

Determine the **wrong** statement about model-averaging ensemble in neural networks.

- We aim at obtaining a strong learner from a combination of weak learners.
- We train the models using different training samples.
- We train networks with the same configuration but different initial random weights.
- We provide an equal weight to the prediction generated by each model when computing the overall prediction from the ensemble.

Question 14

Consider the following loss function. It computes the average between the absolute error of the first output and the squared error of the second output.

```
def custom_hybrid(b):
  def custom_loss(y_true, y_pred):
    error = tf.math.abs(y_true[:,0] - y_pred[:,0])
    loss1 = tf.where(b > error, 0, error)
    loss2 = tf.square(y_pred[:,1] - y_true[:,1])
    return 0.5 * (loss1 + loss2)
  return custom_loss
```

Determine the **correct** statement.

- The squared error component is equal to zero for those predictions where the absolute error is above a parameter "b."
- The absolute error component is equal to zero for those predictions where the absolute error is above a parameter "b."
- The squared error component is equal to zero for those predictions where the absolute error is below a parameter "b."
- The absolute error component is equal to zero for those predictions where the absolute error is below a parameter "b."

Question 15

Consider the following modification to the Walk Forward loop implemented in lesson 2.

```
n_shift = n_train + 240
for i in range(0, len(y)-n_shift, n_test):
  X_train, X_test = X[i:i+n_train], X[i+n_shift:i+n_shift+n_test]
  y_train, y_test = y[i:i+n_train], y[i+n_shift:i+n_shift+n_test]
```

Determine the right answer.

- It will reduce the likelihood of leakage by excluding 240 observations from the training set that are contiguous to the test set.

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- It will eliminate leakage by excluding 240 observations from the training set that are contiguous to the test set.
- It will reduce the number of observations in each training set.
- (•) It will reduce the likelihood of leakage by excluding 240 observations where the training sample and the test sample overlap.

Question 16

In the Combinatorial Purged Cross-Validation, N refers to which of the following?

- The number of backtest paths that we can generate
- The number of groups that we include in each test sample
- The number of observations that compose the time series
- (•) The number of groups into which we split the time series

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